MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 8334

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

DESCRIPTION: Triode, miniature

Outline	---	5-1 (EIA)
Base	---	E7-1, silver-plated pins
Envelope	---	T5-1/2
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5	6	7
Element	---	a	g	h	h	k	g	a

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ik	Pp	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	W	°C	ft
Maximum:	6.9	330	0	100	---	1.0	33	4.4	125	Note 1
Minimum:	5.7	---	---	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	200	0	0	100	---	---	---	---	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1							
1266	Total grid current	Eb = 250 Vdc; Rk = 250 ohms (see note 2)	0.65	II	Ic	0	-1.5	μAdc
1256	Electrode current (1) (anode)		0.65	II	Ib	14	22	mAdc
1306	Transconductance (1)		0.65	II	Sm	6,000	13,500	μmhos
1201	Short and discontinuity detection		0.4	II	---	---	---	---
	Quality conformance inspection, part 2							
1336	Heater-cathode leakage		4.0	I	Ihk	---	10	μAdc
1301	Heater current		4.0	II	If	205	245	mA
1256	Electrode current (2)(anode)	Ec = -10 Vdc; Rk = 0	4.0	I	Ib	---	75	μAdc
1211	Insulation of electrodes		4.0	S3	R	100	---	Meg
1306	Transconductance (2)	Ef = 5.7 V	4.0	I	ΔSm Ef	---	15	%
(C) 1246	Audio frequency noise	Ebb = 200 Vdc; Ec1 = -2.5 Vdc; Rk = 0; Ecal = 60 mVac; Rg = 0.1 Meg; Rp = 10,000 ohms (see note 3)	2.5	I	---	---	---	---
1316	Amplification factor		4.0	Code F	Mu	40	70	---
1331	Direct-interelectrode capacitance	Shield 316	6.5	Code E	Cgp	1.3	2.1	pF
					Cin	2.5	4.1	pF
					Cout	---	2.3	pF
					Ck-g+h	3.5	6.9	pF
					Cp-g+h	2.6	3.4	pF
1031	Low-frequency vibration	Rp = 2,000 ohms	2.5	I	Ep	---	50	mVac
1041	Shock	450 G (see note 4)	---	---	---	---	---	---
1031	Vibration-fatigue test	2.5 G; F = 25 min 60 max; fixed frequency	6.5	See note 5	---	---	---	---
---	Post-shock and vibration-fatigue test end points	Low-frequency vibration	---	---	Ep	---	100	mVac
		Heater-cathode leakage	---	---	Ihk	---	20	μAdc
		Change in transconductance (1) of individual tubes	---	---	ΔSm t	---	20	%
		Total grid current	---	---	Ic	0	-3.0	μAdc
1121	Base strain		---	---	---	---	---	---
2126	Glass strain		2.5	I	---	---	---	---
1105	Permanence of marking		---	---	---	---	---	---

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METHOD	REQUIREMENT OR TEST	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSP LEVEL OR CODE	ALLOWABLE DEFECTIVES PER CHARACTERISTIC	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3								
1506	Heater-cycling life	Ef = 7.5 V; Ehk = 140 Vac; Eb = Ec = 0	---	---	---	---	---	---	---
1521	Survival-rate life	Ehk = -200 Vdc; Rg = 1.0 Meg (see note 6)	---	---	---	---	---	---	---
---	Survival-rate life-test end points	Total grid current	---	---	5	Ic	0	-1.5	µAdc
		Electrode current (1) (anode)	---	---	5	Ib	11	---	mAdc
1501	Intermittent life	Ehk = -200 Vdc; Rg = 1.0 Meg (see note 7)	---	---	---	---	---	---	---
---	Intermittent life-test end points (500 hours)	Inoperatives	---	---	1	---	---	---	---
		Total grid current	---	---	1	Ic	0	-4.5	µAdc
		Electrode current (1) (anode)	---	---	1	Ib	11	---	mAdc
		Total defectives	---	---	2	---	---	---	---
---	Intermittent life-test end points (1,000 hours)	Inoperatives	---	---	1	---	---	---	---
		Total grid current	---	---	2	Ic	0	-4.5	µAdc
		Electrode current (1) (anode)	---	---	2	Ib	11	---	mAdc
		Total defectives	---	---	3	---	---	---	---

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. This test to be performed at the conclusion of the holding period.
3. The rejection level shall be set at the VU meter reading obtained during calibration.
4. A grid resistor of 0.1 Meg shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
5. This test shall be conducted on the initial lot and thereafter on a lot approximately every 6 months. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot passes. When one lot has passed, the 6-month rule shall apply. MIL-STD-105, sample size code letter E, shall apply.
6. Sample size shall be 100 tubes per lot.
7. Sample size shall be 10 tubes per lot.

Custodians:
Army - EL
Navy - EC
Air Force - 80

Review activities:
Air Force - 11, 17
DSA - ES

User activities:
Army - WC
Navy - AS, MC, CG, SH
Air Force - 19

Preparing activity: Navy - EC

Agent: DSA - ES

(Project 5960-2523)

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NAVAL ELECTRONIC SYSTEMS COMMAND
WASHINGTON, D. C. 20360

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

S/N-0102-014-1801 C-25254

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